ORDER-INFORMED DOMAIN GENERALIZATION FOR ROTATING-MACHINERY FAULT DIAGNOSIS

A Comprehensive Research Project Report

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Abstract

This study investigates whether speed-informed signal normalization improves rotating-machinery fault classification under unseen operating conditions and whether a consistency-learning objective provides an additional benefit. The work uses selected recordings from the Paderborn Bearing DataCenter and compares three one-dimensional convolutional neural network (CNN) approaches: a CNN trained on raw vibration windows (Raw CNN), a CNN trained on order-domain windows (Order CNN), and an order-domain CNN trained with an additional consistency objective (Order CNN + Consistency). The evaluation uses four recording-level held-out-condition folds. Each fold contains 288 training recordings, 72 validation recordings, and 120 test recordings. The selected dataset comprises 480 recordings, with 240 labelled healthy and 240 labelled damaged.

Across the four folds, the Raw CNN and Order CNN + Consistency achieved the highest mean accuracy (95.83% ± 8.33%), macro F1 (95.71% ± 8.57%), and balanced accuracy (95.83% ± 8.33%). The Order CNN achieved lower mean accuracy (89.17% ± 21.67%) and macro F1 (86.66% ± 26.68%). In the most challenging held-out-speed fold, the Order CNN achieved 56.67% accuracy and 13.33% damaged-class recall, while the consistency-enhanced order model achieved 83.33% accuracy and 66.67% damaged-class recall. The Raw CNN also achieved 83.33% accuracy and 66.67% damaged-class recall in that fold. These results indicate that the consistency objective substantially improved the order-domain model in the challenging fold, but did not outperform the raw-signal model overall.

The findings are interpreted cautiously because the four folds test unseen operating conditions while bearing identities may appear in training recordings under other conditions. The study therefore evaluates held-out-condition generalization, not generalization to entirely unseen physical bearing units. In addition, the implementation sets the vibration sampling rate to 64 kHz; the selected channel's actual raster must be confirmed against the source MATLAB metadata before the order-domain results can be treated as fully validated. This report records that verification requirement explicitly rather than assuming it has been completed.

Keywords

Rotating machinery; bearing fault diagnosis; domain generalization; order tracking; speed-informed normalization; convolutional neural network; consistency learning; Paderborn Bearing DataCenter.

1. Introduction

1.1 Background

Rotating machinery is widely used in industrial production, transportation, energy systems, and manufacturing. Bearings are critical components in many rotating systems, and bearing degradation can lead

to vibration, efficiency loss, unplanned downtime, and secondary equipment damage. Condition monitoring systems use measured signals, particularly vibration, to identify abnormal behaviour and support maintenance decisions. Automated fault diagnosis is therefore an important application of machine learning in predictive maintenance.

Deep learning methods can learn discriminative patterns directly from measured signals. However, the vibration patterns associated with a fault can change when rotational speed, load, or other operating conditions change. A model trained under one set of conditions may consequently perform less reliably under a condition that was not represented during training. This issue is closely related to domain generalization: the objective is to learn representations that remain useful when the test data come from a different operating domain.

1.2 Problem Statement

A conventional vibration classifier may learn features that are strongly associated with the operating conditions present in its training data. Changes in rotational speed can shift characteristic frequencies and alter the temporal appearance of vibration patterns. As a result, a classifier may perform well on familiar conditions but degrade when evaluated at an unseen speed or a different combination of speed, load, and torque.

Order-domain processing offers a physically motivated way to represent rotating-machine signals. Instead of describing the signal only by time, angular resampling represents it relative to shaft rotation. This can reduce the direct dependence of periodic vibration patterns on rotational speed. Nevertheless, order-domain processing does not automatically guarantee improved classification, particularly when signal metadata, sampling rates, or speed measurements are inaccurate.

1.3 Research Question

Does speed-informed signal normalization improve fault classification at unseen rotational speeds compared with raw vibration-signal learning, and does an additional consistency objective provide further improvement?

1.4 Aim and Objectives

The aim is to evaluate a computationally efficient, speed-informed approach to rotating-machinery fault diagnosis under held-out operating conditions.

- Prepare a recording-level dataset using selected healthy and damaged bearing recordings from the Paderborn Bearing DataCenter.
- Generate matched raw-vibration and order-domain window representations using measured rotational-speed information.
- Implement and compare Raw CNN, Order CNN, and Order CNN + Consistency models.

- Use recording-level held-out-condition folds to evaluate performance under operating conditions excluded from training and validation.
- Measure accuracy, precision, recall, F1-score, macro F1, balanced accuracy, and ROC-AUC.
- Examine whether the consistency objective improves the order-domain model and identify limitations that affect interpretation.

1.5 Research Contribution

The project contributes a controlled comparison of raw vibration learning, speed-informed order-domain learning, and order-domain learning with a consistency objective. Its intended methodological contribution is a relatively lightweight pipeline combining physically informed signal normalization with a training objective that encourages similar predictions for clean and mildly augmented versions of the same order-domain input. The experiments separate the contribution of order-domain representation from the additional consistency term.

1.6 Scope

The study is limited to binary classification of selected Paderborn bearing recordings as healthy or damaged. It uses four held-out operating conditions and three CNN configurations. It does not claim to diagnose every possible bearing defect, generalize to every machine type, or demonstrate deployment in a live industrial system. The unit of data splitting and final evaluation is the original recording rather than an individual window.

2. Literature Review

2.1 Deep Learning for Rotating-Machinery Diagnosis

Deep learning approaches, including one-dimensional CNNs, are widely used for machinery fault diagnosis because convolutional filters can learn local patterns from vibration time series. A raw-signal CNN avoids manually designed frequency-domain features and provides a useful baseline. However, learned features can be sensitive to changes in speed and load if the training data do not adequately represent the operating variability expected at test time. This motivates evaluation protocols that hold out operating conditions rather than randomly distributing windows from the same recordings across training and test sets.

2.2 Domain Generalization and Domain Adaptation

Domain generalization (DG) seeks a model that performs on a target domain that was not available during training. Domain adaptation (DA), in contrast, generally uses some information from the target domain, such as unlabelled target samples, to adapt the model. This distinction matters when comparing related machinery-diagnosis papers: a method described as cross-domain diagnosis or source-free adaptation is not automatically a domain-generalization method under a strict unseen-domain protocol.

Xiao et al. (2025) survey domain-generalization research for rotating-machinery fault diagnosis and describe the motivation for learning features that remain robust across changing operating conditions. The survey provides context for the present study's focus on unseen operating conditions and highlights the importance of evaluation design when claiming generalization.

2.3 Recent Methods Relevant to the Study

Li et al. (2025) present multimodal adaptive signal fusion for domain generalization in imbalanced few-shot rotating machinery fault diagnosis. Their work is relevant because it addresses domain shifts and the use of multiple signal modalities. The present project has a narrower scope: it focuses on a single vibration representation and tests whether speed-informed order-domain normalization and a consistency objective are useful under held-out conditions.

Liu et al. (2026) propose a domain-generalization framework based on causal representation learning and causal feature identification. This line of work seeks representations that are less dependent on spurious, domain-specific correlations. The present project does not claim to perform causal discovery or causal representation learning. Instead, it tests a physically motivated transformation based on measured shaft speed and an augmentation-consistency training objective.

Other recent studies address related but not identical problems. Research on cross-domain diagnosis under time-varying speed and source-free domain adaptation can provide useful context for speed-aware preprocessing, but adaptation methods should be distinguished from the present held-out-condition design. The comparison in this report is therefore conceptual rather than a claim that all cited approaches solve the same learning problem.

2.4 Research Gap and Positioning

The reviewed literature shows substantial interest in robust machinery diagnosis under changing operating conditions. More complex solutions may involve multimodal fusion, causal representation learning, or adaptation to target-domain data. The present project investigates a more compact question: whether measured-speed-informed order-domain normalization improves performance relative to raw-signal learning, and whether a consistency term adds value once order-domain inputs are used. The project is positioned as a controlled empirical comparison, not as a claim that order tracking or consistency learning is entirely new.

3. Methodology

3.1 Research Design

The study follows a quantitative experimental design. Three CNN configurations are trained and evaluated using the same selected recording set and four held-out operating-condition folds. The raw and order-domain models use the same general CNN architecture, while the consistency model adds a second training view and

a consistency penalty. Test predictions are aggregated from windows to the original recording level before calculating metrics.

3.2 Dataset and Label Selection

The dataset is sourced from the Paderborn University Bearing DataCenter. The selected subset contains 480 recordings: 240 healthy recordings and 240 damaged recordings. Healthy bearing identifiers are K001, K002, and K003 (80 recordings each). Damaged identifiers are KA04, KI04, and KA15 (80 recordings each). K004 is excluded from the selected subset. Labels are binary: healthy = 0 and damaged = 1.

The Paderborn Bearing DataCenter provides bearing measurement data and operating information relevant to condition monitoring. The dataset includes vibration-related measurements and operating variables such as rotational speed, torque, and radial force, depending on the measurement configuration. The exact signal channel and its sampling metadata must be retained when deriving order-domain representations.

3.3 Operating Conditions and Fold Design

Fold	Held-out condition	Purpose / interpretation
1	N09_M07_F10	Lower-speed condition (approximately 900 RPM) held out against training conditions around 1500 RPM; primary unseen-speed test.
2	N15_M01_F10	Holds out the M operating-condition factor.
3	N15_M07_F04	Holds out the F operating-condition factor.
4	N15_M07_F10	Holds out a combination of operating-condition factors.

For each fold, the held-out condition contributes 120 test recordings. The remaining 360 recordings are split into 288 training and 72 validation recordings. The split is performed at recording level before window generation is used for model fitting. This prevents windows from the same original recording from being distributed across training, validation, and test sets within a fold.

An important qualification is that the same physical bearing identifiers can occur in training recordings under other operating conditions and in test recordings under the held-out condition. Therefore, the experiment tests generalization to held-out operating conditions for bearing identities represented in the training data. It does not establish generalization to entirely unseen bearing units.

3.4 Signal Processing and Window Generation

Two input representations are prepared. The raw representation consists of vibration windows of 20,000 samples with a stride of 10,000 samples. The order-domain representation is generated by integrating measured rotational speed to estimate angular position, interpolating the vibration signal onto a uniform angular grid, and extracting windows of 2,048 samples. Each order window represents eight revolutions at 256 samples per revolution, with a stride of 1,024 samples.

The order transformation uses the measured rotational-speed series. In simplified form, angular phase is estimated by accumulating angular increments derived from rotational speed and the sampling interval. The

vibration signal is then interpolated at uniformly spaced angular positions. This converts a time-sampled signal into an angle/order-related representation.

3.5 Sampling-Rate Verification Requirement

The experiment script defines VIBRATION_SAMPLING_RATE_HZ = 64,000 Hz and includes a code comment stating that this value must agree with the selected channel's raster. However, the selected channel's actual raster was not independently confirmed during the final location audit. An earlier inspection of a representative MATLAB file showed a channel described as Mech_4kHz, but that observation alone does not prove that this was the channel selected by the final pipeline. If the selected vibration channel was sampled at 4 kHz, using 64 kHz in the angular-phase calculation would be incorrect. Conversely, if the selected channel was the 64 kHz vibration channel, the configured value may be appropriate. The selected channel name and its MATLAB raster metadata must therefore be checked before final submission or before treating the order-domain results as fully validated.

3.6 Normalization and Data Leakage Control

Normalization statistics are computed separately for each fold and representation using only the training windows supplied to the training procedure. The resulting training mean and standard deviation are then applied to training, validation, and test windows for that fold. This design prevents information from validation or test data from influencing the normalization statistics.

3.7 CNN Architecture

The CNN backbone uses one-dimensional convolutional layers to learn local signal patterns. The documented architecture comprises a 32-filter convolution with kernel size 15 and stride 4, batch normalization and max-pooling; a 64-filter convolution with kernel size 9, batch normalization and max-pooling; and a 128-filter convolution with kernel size 7, batch normalization and global average pooling. The learned representation is passed through a 64-unit ReLU dense layer, dropout of 0.4, and a sigmoid output for binary classification.

Component	Configuration
Convolution block 1	Conv1D: 32 filters, kernel 15, stride 4; batch normalization; max-pooling 4.
Convolution block 2	Conv1D: 64 filters, kernel 9; batch normalization; max-pooling 4.
Convolution block 3	Conv1D: 128 filters, kernel 7; batch normalization; global average pooling.
Classifier	Dense 64, ReLU; dropout 0.4; sigmoid output.

3.8 Consistency-Learning Objective

The consistency model receives a clean order-domain window and a mildly augmented version of the same window. Augmentation is applied during training only and consists of amplitude scaling sampled from 0.9 to 1.1 and Gaussian noise with standard deviation 0.02. The model is trained to classify both views correctly while encouraging their predicted probabilities to remain close.

The implemented objective is:

$$L = 0.5\,[BCE(y, p_{clean}) + BCE(y, p_{aug})] + 0.1 \times MSE(p_{clean}, p_{aug})$$

Here, BCE is binary cross-entropy, MSE is mean squared error between the clean-view and augmented-view predictions, y is the binary label, and the consistency weight is 0.1. This objective encourages prediction stability under the specified mild perturbations; it does not by itself prove invariance to all operating-condition changes.

3.9 Training Configuration

Setting	Configuration
Maximum epochs	40
Optimizer	Adam; initial learning rate 0.001
Early stopping	Monitor validation AUC; patience 8; restore best weights/checkpoint as implemented.
Learning-rate reduction	Monitor validation loss; factor 0.5; patience 4; minimum learning rate 0.000001.
Batch sizes	Raw CNN: 8; Order CNN: 32; Order CNN + Consistency: 16.
Random seed	Base seed 42, with fold-specific seed adjustment (42 + fold number).
Decision threshold	0.5 for binary predictions.

3.10 Recording-Level Prediction and Metrics

For each test recording, the model predicts probabilities for its windows. Window probabilities are averaged to produce one recording-level probability. A threshold of 0.5 converts this probability into a binary prediction. Metrics are calculated at recording level, so the number of test predictions is based on recordings rather than the larger number of overlapping windows.

The reported metrics are accuracy, damaged-class precision, damaged-class recall, damaged-class F1, macro F1, balanced accuracy, and ROC-AUC. Accuracy measures the proportion of correct predictions. Precision measures the proportion of predicted damaged recordings that are truly damaged. Recall measures the proportion of damaged recordings detected. F1 is the harmonic mean of precision and recall. Macro F1

averages class-specific F1 scores. Balanced accuracy averages sensitivity across the two classes. ROC-AUC measures ranking discrimination across thresholds.

3.11 Integrity Checks

The project included checks of recording-level split integrity, raw/order window-manifest integrity, prediction integrity, held-out-condition membership, and normalization-statistics provenance. The reviewed audit outputs reported no train/validation/test recording overlap within folds, no duplicate or conflicting manifest rows, matched raw/order window identities, complete expected test predictions, valid probability ranges, and no training or validation recordings from each fold's held-out condition. These checks support the integrity of the data-handling and evaluation pipeline, subject to the unresolved sampling-rate verification described above.

4. Results

4.1 Overall Four-Fold Performance

Table 4.1 reports the mean and standard deviation across the four held-out-condition folds. Values are shown as percentages. The results show that the Raw CNN and Order CNN + Consistency achieved identical mean threshold-based accuracy and macro F1, while the unregularized Order CNN performed less consistently.

Model	Accuracy	Damaged precision	Damaged recall	Damaged F1	Macro F1	Balanced accuracy	AUC
Order CNN	89.17 ± 21.67	100 ± 0	78.33 ± 43.33	80.88 ± 38.24	86.66 ± 26.68	89.17 ± 21.67	94.70 ± 10.60
Order CNN + Consistency	95.83 ± 8.33	100 ± 0	91.67 ± 16.67	95.00 ± 10.00	95.71 ± 8.57	95.83 ± 8.33	99.26 ± 1.49
Raw CNN	95.83 ± 8.33	100 ± 0	91.67 ± 16.67	95.00 ± 10.00	95.71 ± 8.57	95.83 ± 8.33	100.00 ± 0.00

Note: all metrics are expressed as percentages. The values are the four-fold summaries supplied by the experiment outputs. The standard-deviation convention should be stated consistently with the aggregation code or spreadsheet used to produce the summary before formal publication.

4.2 Challenging Held-Out-Speed Fold

Fold 1 holds out N09_M07_F10, the lower-speed condition (approximately 900 RPM) compared with the approximately 1500 RPM conditions used in the other operating settings. This fold is particularly relevant to the research question because it tests a substantial speed change.

Model	Accuracy	Damaged precision	Damaged recall	Damaged F1	Macro F1	Balanced accuracy	AUC
Raw CNN	83.33	100.00	66.67	80.00	82.86	83.33	100.00
Order CNN	56.67	100.00	13.33	23.53	46.65	56.67	78.81
Order CNN + Consistency	83.33	100.00	66.67	80.00	82.86	83.33	97.03

4.3 Fold 1 Confusion Matrices

Model	True negative	False positive	False negative	True positive
Raw CNN	60	0	20	40
Order CNN	60	0	52	8
Order CNN + Consistency	60	0	20	40

All three models correctly classified the 60 healthy test recordings in Fold 1, producing no false positives. The difference was in detection of damaged recordings. The Order CNN detected only 8 of 60 damaged recordings, whereas both the Raw CNN and Order CNN + Consistency detected 40 of 60. The consistency objective therefore improved damaged-class recall for the order-domain model from 13.33% to 66.67% in this fold.

4.4 Folds 2–4

For folds 2, 3, and 4, all three models achieved 60 true negatives, zero false positives, zero false negatives, and 60 true positives in each fold. Thus, all models achieved perfect threshold-based classification in these three held-out-condition folds. This creates a ceiling effect in the aggregate results and means that most of the observed differences across models are driven by Fold 1.

4.5 Interpretation of AUC and Threshold Metrics

The Raw CNN achieved an AUC of 100% in Fold 1 even though its threshold-based accuracy was 83.33%. This is possible because AUC evaluates the ranking of positive and negative examples over all thresholds, whereas accuracy at the selected threshold evaluates one operating point. The Raw CNN may rank all positive examples above all negative examples while some positive probabilities remain below 0.5. This distinction should be explained rather than treating AUC and accuracy as interchangeable.

5. Discussion

5.1 Does Order-Domain Normalization Improve Unseen-Speed Classification?

The results do not support a general claim that order-domain normalization improves classification over raw vibration learning. Across the four folds, the Raw CNN achieved higher mean accuracy and macro F1 than the unregularized Order CNN. In the challenging held-out-speed fold, the Order CNN performed substantially worse than the Raw CNN. The observed outcome suggests that the particular order-domain preprocessing and model configuration used in this experiment were not sufficient, by themselves, to provide robust unseen-speed classification.

This result is useful rather than a failed experiment: a physically motivated representation should be tested against a raw-signal baseline, not assumed to be superior. Potential explanations include information loss or distortion during interpolation, sensitivity to speed-channel quality, mismatch between the configured and actual vibration sampling rate, and the possibility that raw CNN features captured useful signal characteristics that were weakened by the order transformation. These are plausible explanations to investigate, not conclusions established by the current results.

5.2 Does the Consistency Objective Help?

The consistency objective improved the Order CNN in Fold 1. Damaged-class recall increased from 13.33% to 66.67%, and accuracy increased from 56.67% to 83.33%. This is the clearest positive result for the proposed additional training objective. However, the consistency-enhanced model did not outperform the Raw CNN in the same fold and had slightly lower mean AUC across the four folds (99.26% compared with 100.00%). Therefore, the results support a limited conclusion: consistency training improved the order-domain model's performance in the most challenging fold, but it did not establish an overall advantage over raw-signal learning.

5.3 Why the Fold-Level Results Matter

The four-fold averages alone obscure the main behaviour. Folds 2–4 were perfectly classified by every model, whereas Fold 1 exposed a substantial weakness in the unregularized Order CNN. Reporting fold-level metrics and confusion matrices is therefore important. The high average performance should not be interpreted as evidence that all operating-condition shifts are equally difficult or that the models are uniformly robust.

5.4 Methodological Strengths

- Recording-level splitting was used before window-level training, reducing the risk of leakage from overlapping windows of the same recording.
- Training-only normalization statistics were applied to validation and test data.

- Raw and order-domain representations were compared using matched recordings and a common CNN backbone.
- Predictions were aggregated to the recording level before final metrics were calculated.
- Multiple metrics were reported, including class-sensitive measures and ROC-AUC.
- Integrity audits checked split membership, window manifests, prediction completeness, labels, and probability ranges.

5.5 Limitations and Threats to Validity

- Sampling-rate verification remains outstanding. The script uses 64 kHz, but the actual raster of the selected vibration channel must be confirmed from the MATLAB metadata. An incorrect sampling rate would invalidate the angular-phase calculation and could affect all order-domain results.
- The folds hold out operating conditions, but bearing identities may be represented in training under other conditions. The results therefore do not demonstrate generalization to entirely unseen physical bearings.
- The study uses a selected subset of the Paderborn dataset and binary healthy/damaged labels. It does not establish performance across all bearing types, damage severities, or machine platforms.
- Only four held-out operating-condition folds were evaluated. The results may be sensitive to the selected conditions and the chosen recordings.
- The reported four-fold means are based on a small number of condition-level test groups. Fold-level variation is substantial for the Order CNN.
- The augmentation policy is limited to amplitude scaling and additive Gaussian noise. It does not cover all realistic sensor, load, speed, or environmental changes.
- The current report does not include a controlled ablation over multiple consistency weights or augmentation strengths.
- Runtime and computational cost were planned evaluation considerations, but no verified numerical runtime results are available in the reported result set. They should not be invented or reported as measured findings.
- Hardware details and exact package versions should be taken from the final experiment environment log if they are required by the institution's reporting format.

6. Conclusion

This project examined whether speed-informed order-domain normalization and a consistency-learning objective could improve rotating-machinery fault diagnosis under held-out operating conditions. Three CNN configurations were compared using 480 selected Paderborn recordings and four recording-level held-out-condition folds.

The results show that the Raw CNN and Order CNN + Consistency achieved the strongest overall mean accuracy (95.83%) and macro F1 (95.71%). The unregularized Order CNN had lower overall performance and a marked weakness in the held-out-speed fold. Adding the consistency objective substantially improved that order-domain model in Fold 1, increasing damaged-class recall from 13.33% to 66.67%. Nevertheless, the consistency-enhanced model did not exceed the Raw CNN overall. The research question is therefore answered with a qualified finding: the tested order-domain normalization did not demonstrate a general advantage over raw-signal learning, while the consistency objective improved the order-domain model under the most challenging held-out-speed condition.

The strongest practical lesson is that speed-informed transformations and consistency objectives should be evaluated through controlled held-out-condition experiments and compared against raw-signal baselines. Before the order-domain conclusions are finalized, the selected vibration channel's sampling rate must be verified against the source data metadata. The study's generalization claim should also remain limited to held-out operating conditions for the selected bearing identities.

7. Recommendations and Future Work

- Verify and document the exact vibration channel selected from each MATLAB recording, including its raster/sample rate, and confirm that the value used in angular resampling matches that metadata.
- If the sampling rate was incorrect, regenerate order-domain windows and rerun the Order CNN and Order CNN + Consistency experiments before making final claims about those models.
- Add a separate bearing-identity holdout experiment to evaluate generalization to physical bearing units that are entirely absent from training.
- Evaluate additional speed values and operating-condition combinations, preferably with repeated seeds or repeated group splits.
- Test consistency weights and augmentation strengths using a validation-only selection procedure.
- Investigate whether interpolation, angular phase integration, and speed measurement quality contribute to the Fold 1 performance gap.
- Report measured runtime and memory use if computational efficiency is part of the claimed contribution.
- Retain fold-level metrics and confusion matrices alongside aggregate summaries so that difficult operating conditions remain visible.

References

Li, S., Liu, J., Ma, H., & Yang, N. (2025). Multimodal adaptive signal fusion for domain generalization in imbalanced few-shot rotating machinery fault diagnosis. IEEE Transactions on Instrumentation and Measurement, 74. https://doi.org/10.1109/TIM.2025.3643024

Liu, Y., Cheng, W., Wen, W., Fang, Q., & Wu, H. (2026). A novel domain generalization framework for fault diagnosis of rotating machinery based on causal representation learning and causal feature

identification. Engineering Applications of Artificial Intelligence, 164, 113201. https://doi.org/10.1016/j.engappai.2025.113201

Paderborn University, Chair of Design and Drive Technology. (n.d.). Bearing DataCenter: Data sets and download. https://mb.uni-paderborn.de/en/kat/research/bearing-datacenter/data-sets-and-download

Xiao, Y., Shao, H., Yan, S., Wang, J., Peng, Y., & Liu, B. (2025). Domain generalization for rotating machinery fault diagnosis: A survey. Advanced Engineering Informatics, 64, 103063. https://doi.org/10.1016/j.aei.2024.103063

Appendix A. Summary of Experiment Configuration

Item	Recorded configuration
Dataset size	480 recordings: 240 healthy and 240 damaged.
Bearing IDs	Healthy: K001, K002, K003; damaged: KA04, KI04, KA15; K004 excluded.
Fold sizes	Per fold: 288 train, 72 validation, 120 test recordings.
Raw windows	20,000 samples; stride 10,000.
Order windows	2,048 samples; 8 revolutions; 256 samples/revolution; stride 1,024.
Window manifest	39,396 matched window records.
Prediction output	1,440 recording-level predictions across 3 models and 4 folds.
Models	Raw CNN; Order CNN; Order CNN + Consistency.
Threshold	0.5.
Consistency augmentation	Amplitude scaling 0.9–1.1; Gaussian noise standard deviation 0.02.
Consistency coefficient	0.1.

Appendix B. Reproducibility and Submission Notes

The experiment source code and notebook should be retained in the associated GitHub repository together with the consolidated fold-metrics and recording-level prediction CSV files. The notebook should be saved with executed outputs where possible. The raw Paderborn dataset should not be uploaded to a public repository unless the applicable dataset terms and repository size constraints have been checked. The README should describe the dataset source, the expected directory layout, the main experiment entry point, and the steps needed to reproduce the reported results.

Before submission, confirm that the Word report's author/institution details match the required university format; verify the sampling-rate issue; confirm the standard-deviation convention used in the four-fold summary; and ensure that the final notebook and CSV files in GitHub correspond to the results reported in this document.